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Lab 10- Implement Triggers:

```
CREATE DATABASE IF NOT EXISTS inventory_db;
USE inventory_db;

DROP TABLE IF EXISTS AuditLog;
DROP TABLE IF EXISTS Products;

CREATE TABLE Products (
    product_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    stock INT,
    price DECIMAL(10, 2)
);

CREATE TABLE AuditLog (
    log_id INT PRIMARY KEY AUTO_INCREMENT,
    action VARCHAR(50),
    product_id INT,
    log_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

DELIMITER //

CREATE TRIGGER BeforeInsertProduct
BEFORE INSERT ON Products
FOR EACH ROW
BEGIN
```

```
    IF NEW.price < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Price cannot be negative';
    END IF;
    IF NEW.stock < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Stock cannot be negative';
    END IF;
END //

CREATE TRIGGER AfterInsertProduct
AFTER INSERT ON Products
FOR EACH ROW
BEGIN
    INSERT INTO AuditLog (action, product_id)
    VALUES ('INSERT', NEW.product_id);
END //

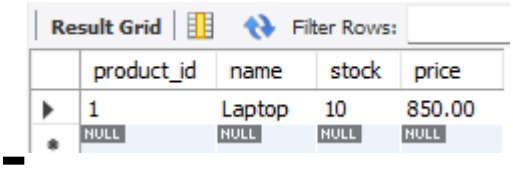
CREATE TRIGGER BeforeUpdateProduct
BEFORE UPDATE ON Products
FOR EACH ROW
BEGIN
    IF NEW.price < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Updated price cannot be negative';
    END IF;
    IF NEW.stock < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Updated stock cannot be negative';
    END IF;
END //

CREATE TRIGGER AfterUpdateProduct
AFTER UPDATE ON Products
FOR EACH ROW
BEGIN
    INSERT INTO AuditLog (action, product_id)
    VALUES ('UPDATE', NEW.product_id);
END //
```

```
CREATE TRIGGER BeforeDeleteProduct
BEFORE DELETE ON Products
FOR EACH ROW
BEGIN
   IF OLD.stock > 0 THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT = 'Cannot delete a product with existing stock';
   END IF;
END //

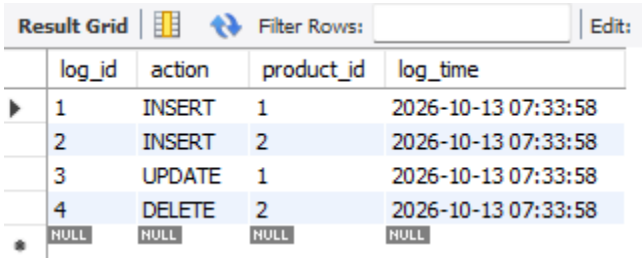
CREATE TRIGGER AfterDeleteProduct
AFTER DELETE ON Products
FOR EACH ROW
BEGIN
   INSERT INTO AuditLog (action, product_id)
   VALUES ('DELETE', OLD.product_id);
END //

DELIMITER ;
```



Result Grid | Filter Rows:

product_id	name	stock	price
1	Laptop	10	850.00
NULL	NULL	NULL	NULL



Result Grid | Filter Rows: | Edit:

log_id	action	product_id	log_time
1	INSERT	1	2026-10-13 07:33:58
2	INSERT	2	2026-10-13 07:33:58
3	UPDATE	1	2026-10-13 07:33:58
4	DELETE	2	2026-10-13 07:33:58
NULL	NULL	NULL	NULL